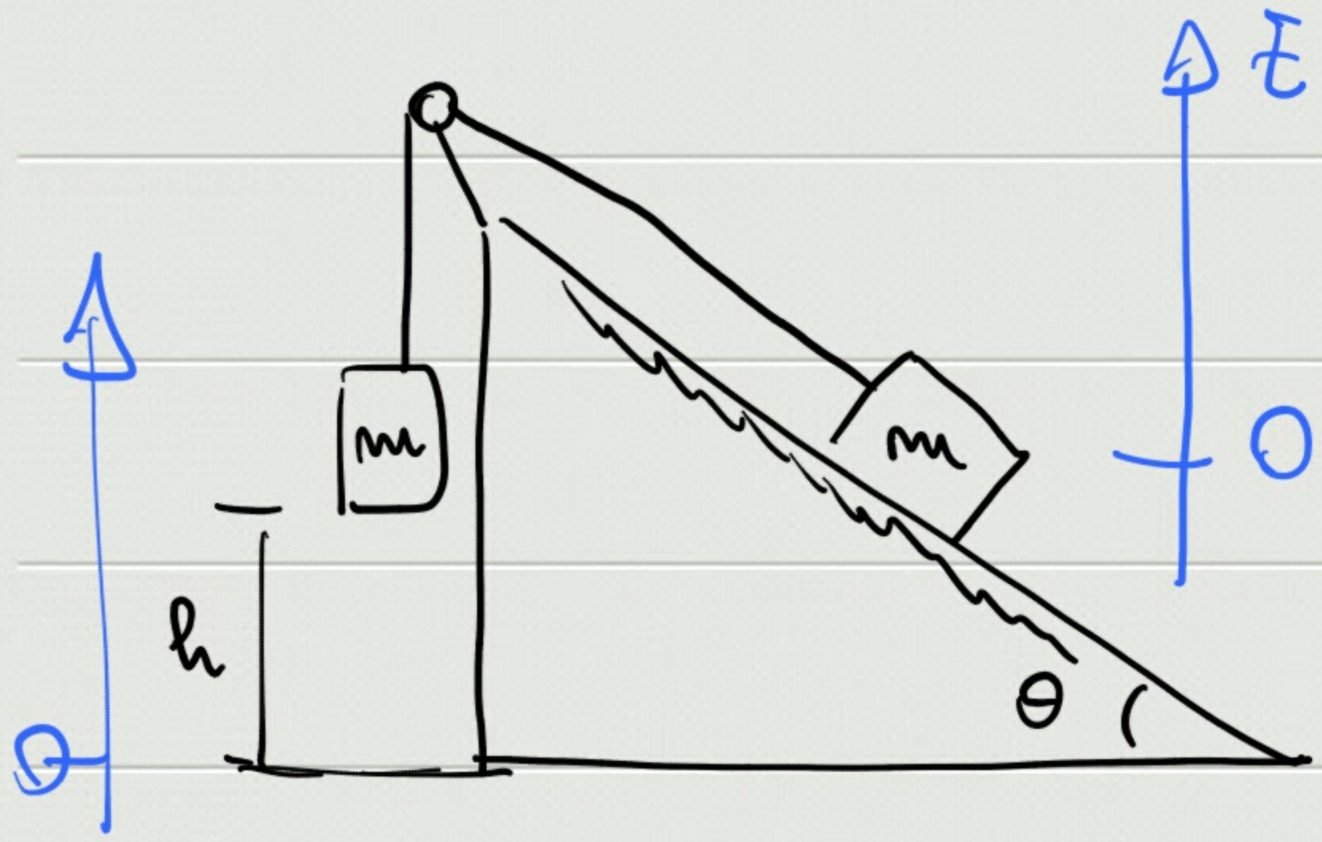
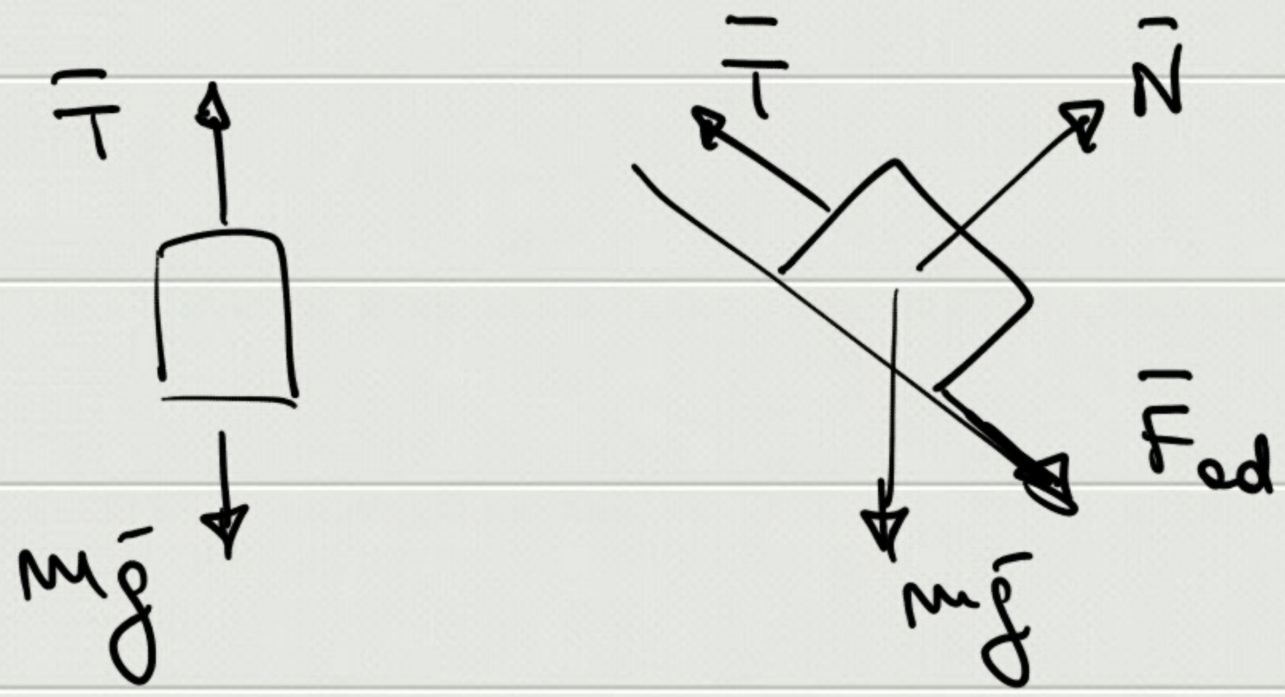


$\theta = 30°$

$\mu_d = 0.4$

$h = 1\ m$

$v_{01} = v_{02} = 0$



$\Rightarrow a$

$\Rightarrow v^2 = 2ah$

$$\frac{1}{2} m v^2 = mgh$$

$$\frac{1}{2} m v^2 + mgh \sin\theta = 0$$

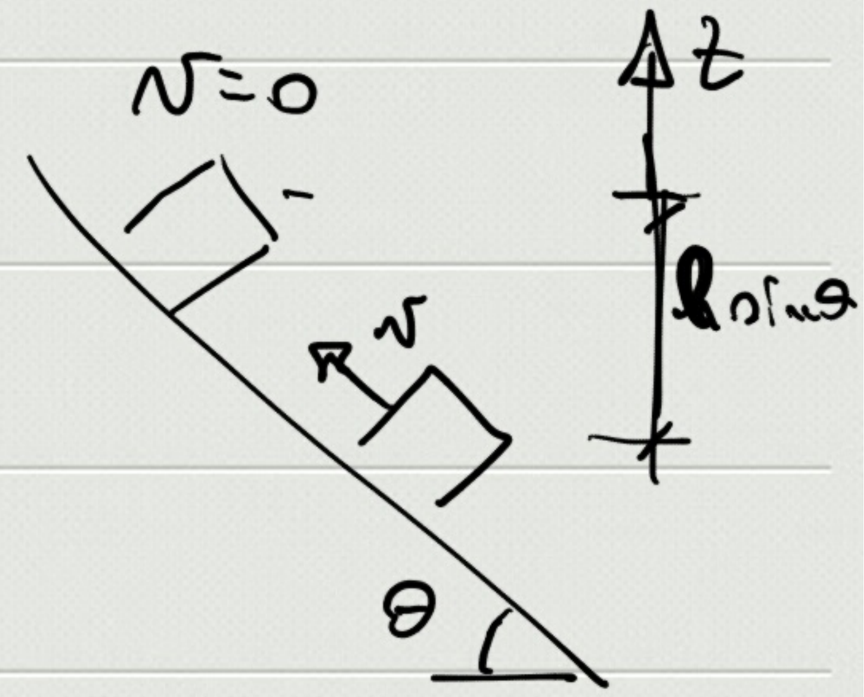
$$m v^2 + mgh \sin\theta = mgh$$

$$-\mu \cancel{m} g \cos\theta\, h = \cancel{m} v^2 + \cancel{m} gh \sin\theta - \cancel{m} gh \quad *$$

$$W_{nc} = \Delta E_m$$

$$v^2 = gh(1 - \sin\theta - \mu\cos\theta)$$

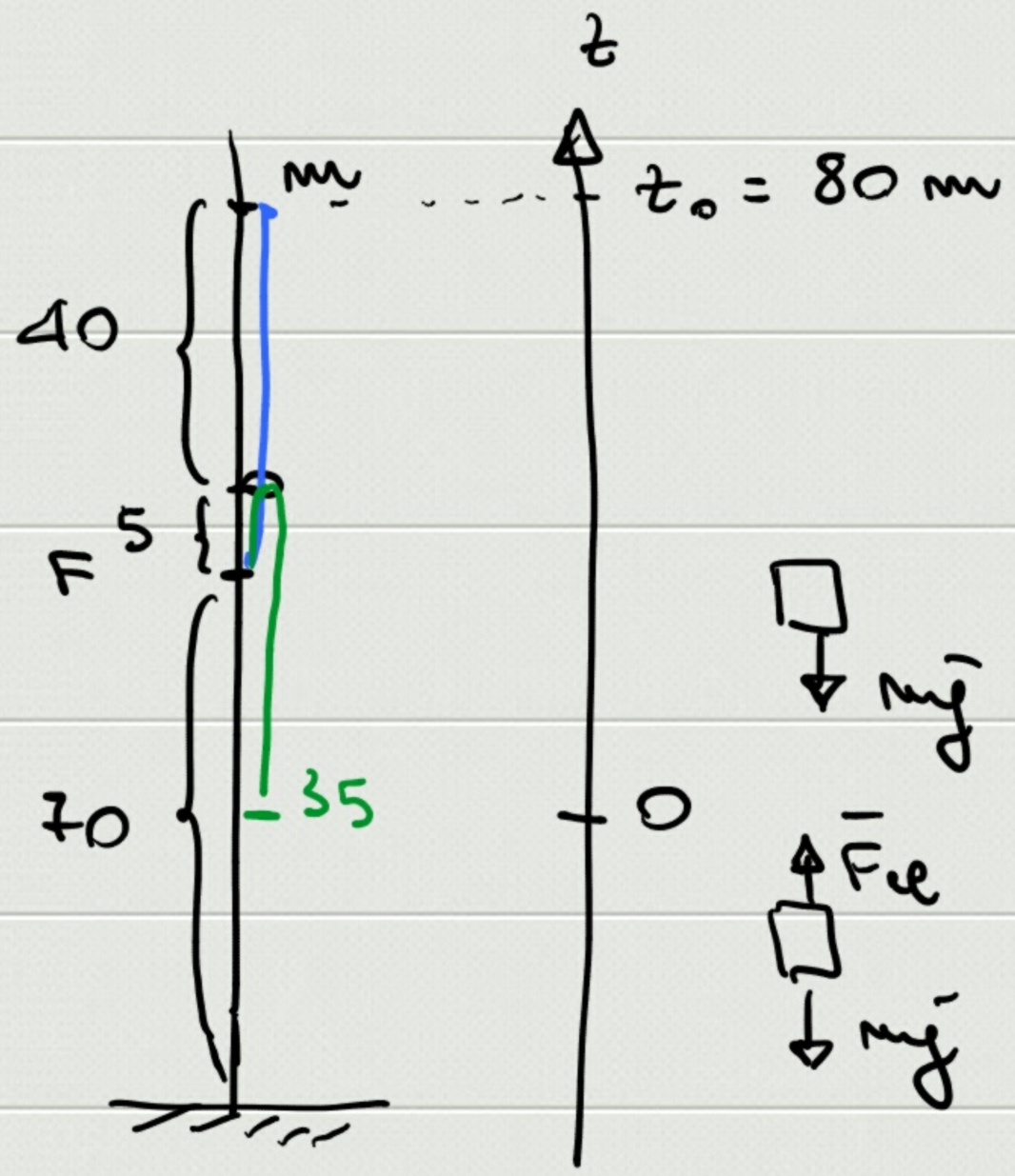
$$\Rightarrow v = 1.23\ m/s$$



$$W_{nc} = \Delta E_m$$

$$-\mu_d \cancel{m} g \cos\theta \cdot l = \cancel{m} g l \sin\theta - \frac{1}{2} \cancel{m} v^2$$

$$l = \frac{v^2}{2g(\sin\theta + \mu\cos\theta)} = 0.091 \text{ m}$$



$m = 80$ kg

$k = 520$ N/m

$\Delta z_{max} = ?$

$v_{max} = ?$ $z(v_{max}) = ?$

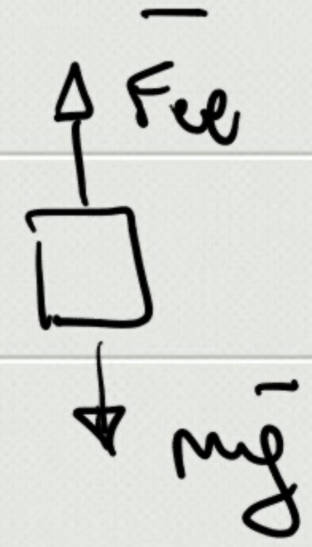
$z \geq 0$ $\quad mgz_0 = \frac{1}{2} m v^2(z) + mgz$

$z < 0$ $\quad mgz_0 = \frac{1}{2} m v^2(z) + mgz + \frac{1}{2} k z^2$ $\quad (*)$

z_{max} : $v(z_{max}) = 0 \Rightarrow z^2 + \frac{2mg}{k} z - \frac{2mg}{k} z_0 = 0$

$$z = -\frac{mg}{k} \pm \sqrt{\left(\frac{mg}{k}\right)^2 + \frac{2mg}{k} z_0} = -17.1 \text{ m}$$

$\cancel{z = 0}$



$-mg - kz = 0$ $\quad * \Rightarrow \boxed{z = -\frac{mg}{k}}$

$$\frac{d}{dz}\left(mgz_0 - mgz - \frac{1}{2} k z^2\right) = 0$$

$\frac{1}{2} m v^2$

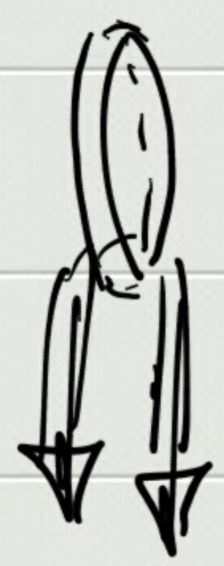
$mg \cancel{- 2kz} = 0$

$T_{rott,fune} = 1.9 \cdot 10^4$ N

$T_{rott,mosch} = 2.2 \cdot 10^4$ N

$T_{max} = |-k z_{max}| < T_{rott}$

$T_{max} = 8892$ N



$T_{mosch,max} = 2 T_{max} \sim 1.8 \cdot 10^4$ N